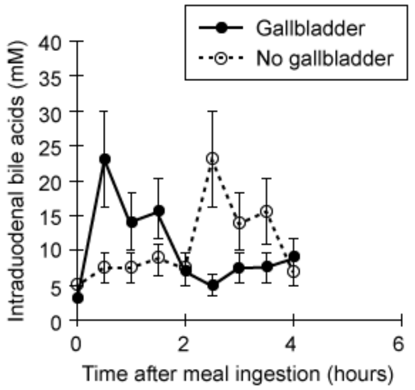
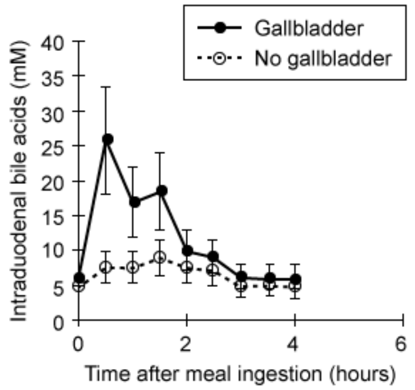
Researchers studying gallbladder function measured intraduodenal bile acid concentrations before and after a standardized meal in two groups of nurses. (The nursing profession was chosen to mitigate lifestyle-related confounding factors.) Nurses in one group had undergone gallbladder removal surgery as patients and were matched for age, sex, and body mass index with nurses in a control group who had not undergone gallbladder removal surgery. Based on this experiment, which of the following graphs best represents how intraduodenal bile acid concentrations in both groups compare after a meal?

○ A.



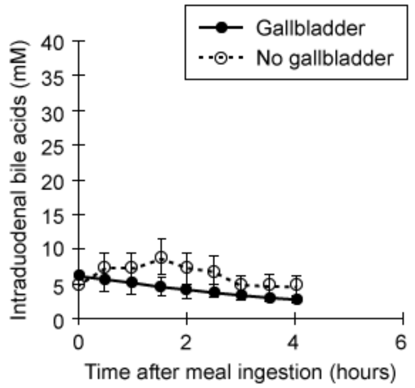
[10%]

✔ ● B.



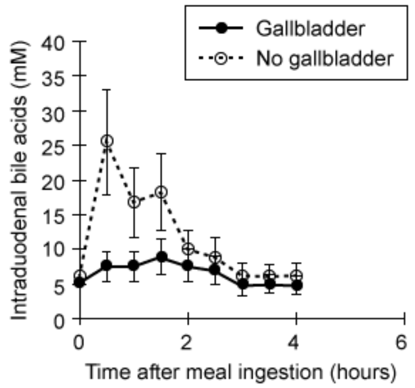
[78%]

○ C.



[1%]

○ D.



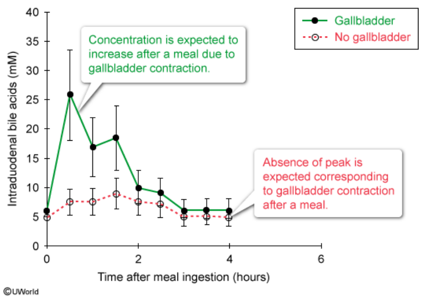
[10%]

Correct
78% answered correctly

微信Mai_ange

Explanation:

The gallbladder stores bile, which is released after meals



Bile is an alkaline fluid that **facilitates fat digestion**; it is **produced by the liver** but **stored in the gallbladder**. Bile contains compounds such as bile acids, cholesterol, and bile pigments. Before they are released from the liver, bile acids are conjugated with additional compounds to form bile salts. Both bile acids and bile salts are amphipathic (containing hydrophobic and hydrophilic regions), a property essential for fat digestion. On a molecular level, the hydrophobic regions of bile salts associate with fat globules whereas the hydrophilic regions associate with the aqueous environment.

Upon meal ingestion, food traverses the mouth and esophagus to enter the stomach, where it is further digested by gastric acid and enzymes to form an acidic, semifluid mixture known as chyme. Chyme then leaves the stomach and enters the **duodenum**, the first segment of the small intestine. In the duodenum, the presence of meal-derived fats within chyme and the acidity of chyme itself stimulate bile release from both the liver and the gallbladder. Bile secretion into the duodenum promotes the neutralization of chyme and the physical digestion of fats (ie, emulsification). In addition, the chemical digestion of fats in the small intestine is carried out by lipases. Fat digestion products (eg, free fatty acids, monoglycerides) are ultimately absorbed via the microvilli of small intestine epithelial cells.

In this scenario, researchers measured intraduodenal bile acids following consumption of a standardized meal. In the **control group** (ie, subjects with a gallbladder), an early **peak** in **intraduodenal bile acid concentration** is **expected** after meal ingestion; this increase corresponds to the release of stored bile from the gallbladder. In contrast, **such a peak** is **absent** in subjects **without a gallbladder**. The bile produced by these subjects is continuously released into the blood from the liver, and therefore does not produce the peak observed when stored bile is released from the gallbladder. As a result, the graph shown in Choice B corresponds to these expected results; the other graphs do not display data consistent with these results.

Educational objective:
Bile is synthesized by the liver to break down (emulsifies) dietary fat. The gallbladder serves as bile storage before food enters the small intestine. In response to meal consumption, bile is released into the small intestine from the liver and from the gallbladder.

Time spent:0 sec
Subject:
Biology

QID:402778
System:
Digestion and Excretion

微信Mai_ange